

The rate expression and its experimental determination (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to deduce the rate equation for a reaction from experimental data.

Some of us use route planner apps to plan out a journey. This tells us how long it will take to get to our destination for various different routes. It also tells us how long it will take if we use a different mode of transport. How does it do that at the touch of a button? Most of these apps work on some kind of mathematical model or equation which is solved by putting in values for different variables, like distance (often calculated automatically from start and end point) and mode of transport. The benefit of equations is that they provide a very precise way to deduce the impact of changing variables on various phenomena in the world.

Google maps, for example, is an extended app that is also capable of accounting for any congestion on the way to give you an estimated arrival time. This means it uses an expanded model that takes into account many other variables like bottlenecking or incidents that may occur across the route.

A rate expression or rate law does exactly that. It is a mathematical equation which calculates the effect of change in concentration of various reactants on the rate of reaction.

In section R2.2.2–3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/>) you learned how a change in concentration of the reactants changes the rate of the equation. A rate equation shows this effect mathematically.

Measuring the rate of reaction

For the purposes of rate equations, the rate of a reaction is measured in terms of how fast the concentration of one of the reactants is falling. Its units are $\text{mol dm}^{-3} \text{s}^{-1}$.

We know that increasing the concentration of the reactants increases the overall rate of the reaction, but not all reactants affect the rate in the same way. Some reactants have a greater influence on the rate of reaction than others. Why do you think that is? Think about reaction mechanisms and rate determining steps. Do all the reactants impact the rate of reaction similarly? In order to understand how reactants can influence the rate of a reaction we need to explore further.

Suppose you carry out an experiment to see how the rate of reaction changes when you alter the concentration of a reactant, A, Some of the possibilities are:

I. The rate of reaction is directly proportional to the concentration of A

This means if you double the concentration of A the rate also doubles. Increasing the concentration three times increases the rate by three times. We can write this as:

$$\text{rate} \propto [\text{A}]$$

II. The rate of reaction is proportional to the square of the concentration of A

If you were to double the concentration of A the rate will go up four times (2^2) or if you increase the concentration of A four times the rate goes up sixteen times ($4^2=16$). We can write this as:

$$\text{rate} \propto [\text{A}]^2$$

III. Extending the possibilities to other reactants

By performing similar experiments for reactant B, in a reaction between A and B, you would see that the reactant concentrations are related to the rate in the following way:

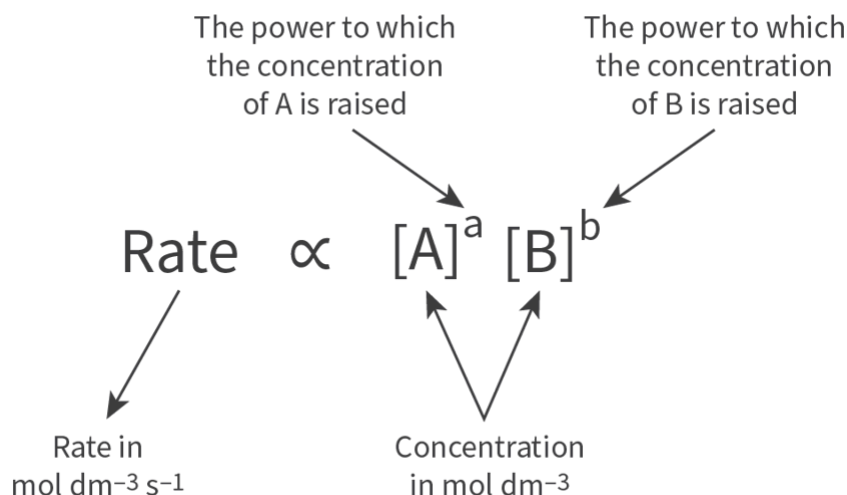


Figure 1. The rate expression for the reaction in which A and B are reacting.

More information for figure 1

The image is a diagram explaining the rate expression for a chemical reaction involving two reactants, A and B. It shows the proportionality relation as "Rate $\propto [\text{A}]^a [\text{B}]^b$ ". This indicates that the rate is proportional to the concentrations

of A and B, raised to their respective powers, a and b. Arrows point out and describe key parts of the equation:

- An arrow pointing to "Rate" is labeled "Rate in mol dm⁻³ s⁻¹", specifying the units for the rate of reaction.
- The concentration of reactant A is labeled as "[A]" and has an accompanying arrow stating "The power to which the concentration of A is raised", indicating its exponent "a".
- Similarly, for reactant B, "[B]" is labeled with an arrow pointing to its exponent, "b", indicating "The power to which the concentration of B is raised".
- Another set of arrows points to both [A] and [B], specifying "Concentration in mol dm⁻³" for these reactants.

[Generated by AI]

If the concentration of a reactant has no effect on the rate, then the concentration of the reactant is raised to a zero. Anything raised to the power of zero is one, this means that the concentration of this reactant does not appear in the rate expression. Therefore, if doubling the concentration of A doubles the rate and changing the concentration of B has no effect on the rate of reaction, the rate expression can be written as:

$$\text{rate} \propto [\text{A}]$$

where the concentration of B is not included in the rate expression.

Moreover, if the rate expression for the reaction between A and B is:

$$\text{rate} \propto [\text{A}][\text{B}]$$

It means doubling the concentration of A will double the reaction rate. The same goes for B, doubling the concentration of B doubles the rate of reaction. That is the rate of reaction is directly proportional to the concentration of both A and B.

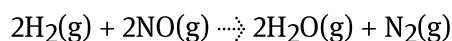
A reaction with the rate expression:

$$\text{rate} \propto [\text{A}]^2$$

means that doubling the concentration of A increases the rate four times. The concentration of B has no effect on the rate of reaction.

Worked example 1

In the reaction:



The rate of reaction doubles when the concentration of H₂ doubles.

Doubling the concentration of NO increases the rate four times.

Write the rate equation for this reaction.

$$\text{rate} \propto [\text{NO}]^2 [\text{H}_2]$$

Determining the rate expression experimentally using the initial rates method

Determining rate expressions experimentally involves determining the effect of changing concentration on rate with respect to each reactant individually. The initial rates method is used and as the name implies, the initial rates of the reactants are used to determine their effect on the rate.

Consider a reaction with two reactants A and B. By measuring the rate of reaction with different initial concentrations of A, while keeping the concentration of B constant, the effect of different concentrations of A can be found. For example, if changing the concentration of A has no effect on the rate it means the order of reaction with respect to A is zero. If the rate doubles when the concentration is doubled it means that the order of reaction is one with respect to A. If doubling the concentration of A results in a four times increase in rate it means the reaction is second order with respect to A.

These steps can then be repeated with different initial concentrations of B, while A is kept constant. This means any change in rate is only due to the concentration of B.

Now try the following worked example to see how initial rate data can be used to deduce the rate expression of a reaction.

Worked example 2

Use the data in the table to work out the rate expression for the reaction between A and B.

Experiment number	Concentration / mol dm^{-3} [A]	Concentration / mol dm^{-3} [B]
1	0.100	0.100
2	0.100	0.200
3	0.200	0.100

Looking at experiment 1 and experiment 2.

Doubling the concentration of B while keeping the concentration of A constant results in no change in the rate of reaction. This means that doubling the concentration of B has no effect on the rate of reaction. Thus the concentration of B will be raised to the power of zero and will not appear in the rate expression.

Looking at experiment 1 and experiment 3.

Doubling the concentration of A while keeping the concentration of B constant results in a four times increase in the rate of reaction. This means that doubling the concentration of A increases the rate four times (2^2). Thus the concentration of A will be raised to the power of two.

Hence the overall rate expression is:

$$\text{rate} \propto [\text{A}]^2$$

Now try a worked example that is a little bit more difficult.

Worked example 3

Use the data in the table to work out the rate expression for the reaction between A and B.

Experiment number	Concentration / mol dm^{-3} [A]	Concentration / mol dm^{-3} [B]
1	0.007	0.004
2	0.014	0.004
3	0.014	0.008

Looking at experiment 1 and experiment 2.

Doubling the concentration of A while keeping the concentration of B constant results in the rate increasing from $2.25 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ to $9 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$.

In order to see how much rate has changed divide the initial rate of experiment 2 by the initial rate of experiment 1:

$$\frac{\text{initial rate experiment 2}}{\text{initial rate experiment 1}} = \frac{9 \times 10^{-4}}{2.25 \times 10^{-4}} = 4$$

This means doubling the concentration of A increases the rate four times. Thus the concentration of A will be raised to the power of 2 in the rate equation.

Looking at experiment 2 and experiment 3.

Doubling the concentration of B while keeping the concentration of A constant results in the rate increasing from $9 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ to $1.8 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$.

In order to see how much rate has changed divide the initial rate of experiment 3 by the initial rate of experiment 2.

Going from experiment 2 to 3, the concentration of B has doubled while the concentration of A remains constant. In order to calculate the factor by which rate has changed:

$$\frac{\text{initial rate experiment 3}}{\text{initial rate experiment 2}} = \frac{1.8 \times 10^{-3}}{9 \times 10^{-4}} = 2$$

This means doubling the concentration of B doubles the rate, hence the rate of reaction is directly proportional to the concentration of B. Thus the concentration of B will be raised to the power of 1 in the rate expression.

Hence the overall rate expression is:

$$\text{rate} \propto [\text{A}]^2[\text{B}]$$

Table 1. A summary of how changing the concentration affects the rate of reaction when the concentration of the reactant is changed.

Changing the concentration of Reactant A	When the concentration of A is raised to the power of 1	When the concentration of A is raised to the power of 2
Doubling [A]	Doubles rate	Rate increases 4×
Tripling [A]	Triples rate	Rate increases 9×
Quadrupling [A]	Quadruples rate	Rate increases 16×

Worked example 4

The reaction $A + B \rightarrow A_2B$ has the rate expression of rate $\propto [A][B]^2$.

What will the effect on the rate of reaction be if the concentration of A is first tripled and then the concentration of B is halved?

$$\text{rate} \propto [A]$$

$$\text{let } [A] = a$$

then

$$\text{rate} \propto 3a$$

Tripling the concentration of A means the rate will increase three times.

$$\text{rate} \propto [B]^2$$

$$\text{Let } [B] = b$$

then

$$\text{rate} \propto \left(\frac{b}{2}\right)^2$$

$$\text{rate} \propto \frac{b^2}{4}$$

Halving the concentration of B means the rate will be reduced to one quarter of its original.

The net effect is the rate of reaction will increase by 0.75 times (3/4).

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Inquirer
 - Thinker
 - Reflective
- **Approaches to learning:**
 - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new contexts
 - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly

- Social skills — Working collaboratively to achieve a common goal
- Time required to complete activity: 45 minutes
- Activity type: Group activity

In this activity you will work in three small groups to simulate the connection between reactant concentration and the rate equation.

Materials

- three small boxes of equal volume
- fifteen red spheres/ping pong balls
- fifteen blue spheres/ping pong balls

Preparation of boxes

Place five red and five blue spheres in box 1.

Place three red and seven blue spheres in box 2.

Place seven red and three blue spheres in box 3.

Alternatively you could carry out the activity by drawing diagrams of the boxes containing red (A) and blue (B) spheres.

Data provided

Each box represents the reaction between A and B. The red spheres represent A and the blue spheres represent B. Since all the boxes are of the same volume the number of spheres in each box represents their concentration.

The rate of reaction for each box is represented as:

$$\text{rate} \propto [\text{A}]^a [\text{B}]^b$$

The rates of reaction in each box are:

Box 1: rate $\propto 125$

Box 2: rate $\propto 147$

Box 3: rate $\propto 63$.

Task

1. Assign one box to each group.
2. In your individual groups, brainstorm using the data provided and the concentration of the reactants in your box to figure out the rate expression for the reaction between A and B.

3. Once everyone has come up with an expression, compare results. Is there any group that does not agree?
4. Remember all three boxes represent the same reaction between A and B. Is it possible to come to a mutual agreement about the rate expression?

The rate expression is $\text{rate} \propto [\text{A}] [\text{B}]^2$